

Week 6: Statistical Foundation

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Contents

Warm-Up	2
Today's Goal	3
Range	3
Mean	3
Median	4
Mode	4
Walkthroughs	5
Common Mistakes	6
Practice Problems	7

※ Warm-Up

A blackboard consists of the numbers

5, 10, 7, 9, 3, 9, 10, 12, 14, 10, 9, 12, 4.

- (a) Sort the numbers from smallest to highest.
- (b) What is the middle number?
- (c) What is the difference between the largest number and the smallest number on the blackboard?
- (d) What number appears the most?
- (e) What is the sum of the numbers on the blackboard?
- (f) How many numbers are on the blackboard?
- (g) Divide the result in (e) by the result in (f). What do you get?

- (a) The sorted list is 3, 4, 5, 7, 9, 9, 9, 10, 10, 10, 12, 12, 14.
- (b) There are 13 numbers, so the middle number is the 7th number. The middle number is 9.
- (c) The largest number is 14 and the smallest number is 3. Their difference is $14 - 3 = 11$.
- (d) The numbers 9 and 10 both appear 3 times, so both appear the most.
- (e) The sum is $5 + 10 + 7 + 9 + 3 + 9 + 10 + 12 + 14 + 10 + 9 + 12 + 4 = 114$.
- (f) There are 13 numbers on the blackboard.
- (g) Dividing the sum by the number of numbers gives $\frac{114}{13} = 8\frac{10}{13} \approx 8.77$.

This warm-up has already introduced the main ideas for today:

- The difference between the largest and smallest number is called the **range** as seen in (c).
- The number in the middle is called the **median** as seen in (b).
- The number that appears the most is called the **mode** as seen in (d).
- The sum divided by how many numbers there are is called the **mean** as seen in (g).

※ Today's Goal

Today we will learn about four important ways to describe a list of numbers:

Word	Meaning
Range	How spread out the numbers are.
Mean	The average, found by adding the numbers and dividing by how many numbers there are.
Median	The middle number after sorting the list.
Mode	The number or numbers that appear the most.

※ Range

The **range** tells us how far apart the largest and smallest numbers are. To find the range:

1. Find the largest number.
2. Find the smallest number.
3. Subtract: Range = Largest Number – Smallest Number.

Find the range of

6, 10, 3, 12, 9.

First, sort: 3, 6, 9, 10, 12. The smallest number is 3 and the largest number is 12, so the range is

$$\text{Largest Number} - \text{Smallest Number} = 12 - 3 = \boxed{9}.$$

※ Mean

The **mean** is also called the **average**.

To find the mean, use the formula

$$\text{Mean} = \frac{\text{Sum of the Numbers}}{\text{Number of Numbers}}.$$

Another way to think about the mean is as a “fair share.” If the total amount were split evenly among all the numbers, the mean is what each number would get.

Find the mean of

4, 8, 10, 14.

First, add the numbers: $4 + 8 + 10 + 14 = 36$. There are 4 numbers, so divide by 4:

$$\text{Mean} = \frac{\text{Sum of the Numbers}}{\text{Number of Numbers}} = \frac{36}{4} = \boxed{9}.$$

Important Note

The mean does not have to be one of the numbers in the list. For example, the mean of 1, 2, 9 is

$$\frac{1 + 2 + 9}{3} = \frac{12}{3} = 4.$$

The number 4 is not in the original list, but it is still the mean.

※ Median

The **median** is the middle number after the numbers are sorted from smallest to largest. To find the median:

1. Sort the numbers from smallest to largest.
2. Find the middle number.

There are two possible situations.

- If there is an odd number of numbers, there is exactly one middle number. Let's consider the following example:

Find the median of

9, 1, 4, 6, 10.

First, sort the numbers:

1, 4, 6, 9, 10.

The middle number is 6, so the median is 6.

- If there is an even number of numbers, there are two middle numbers. In this case, we average the two middle numbers. Let's consider the following example:

Find the median of 1, 2, 3, 4.

The two middle numbers are 2 and 3, so the median is

$$\frac{2 + 3}{2} = \frac{5}{2} = 2.5.$$

Therefore, the median is 2.5.

※ Mode

The **mode** is the number that appears the most. Unlike the median, there can be more than one mode. If there is one mode, usually the problem will state "unique mode". I say usually because sometimes it won't.

Find the mode of

2, 3, 3, 4, 5

For this the problem is already sorted. It is clear this has mode 3 because 3 appears more than any other number.

Find the mode of

1, 1, 3, 5, 5

Once again, this problem has a list which is already sorted. It is clear this has two modes: 1 and 5. Both 1 and 5 appear twice, and no other number appears more often.

✧ Walkthroughs

In three bowling games, Alice scores 139, 143, and 144. What score will Alice need in a fourth game in order to have an average score of 145 for all four games?

We want Alice's average after 4 games to be 145. Since

$$\text{Mean} = \frac{\text{Sum of Scores}}{\text{Number of Scores}},$$

we need

$$\frac{139 + 143 + 144 + s}{4} = 145,$$

where s is Alice's fourth score.

First, find the total score Alice needs after 4 games: $145 \cdot 4 = 580$. Now find the sum of the first three scores:

$$139 + 143 + 144 = 426.$$

So the fourth score must be $580 - 426 = \boxed{154}$.

A number N is inserted into the list

2, 6, 7, 7, 28.

The mean is now twice as great as the median. What is N ?

After inserting N , there will be 6 numbers. The sum of the original numbers is

$$2 + 6 + 7 + 7 + 28 = 50.$$

So the new mean is $\frac{50 + N}{6}$. Since there are 6 numbers, the median will be the average of the 3rd and 4th numbers after sorting.

We need $\text{Mean} = 2 \cdot \text{Median}$. Now we consider cases based on where N fits into the sorted list.

- If $N \leq 6$, then the middle two numbers will be 6 and 7.

So the median is $\frac{6+7}{2} = \frac{13}{2} = 6.5$. The mean must be twice the median: $2 \cdot 6.5 = 13$. So

$$\frac{50 + N}{6} = 13.$$

Multiply both sides by 6 then $50 + N = 78$. Thus, $N = 28$. But this does not satisfy $N \leq 6$, so there is no solution in this case.

- If $6 < N < 7$, then the sorted list looks like

$$2, 6, N, 7, 7, 28.$$

The middle two numbers are N and 7, so the median is $\frac{N+7}{2}$. The mean is twice the median, so

$$\frac{50 + N}{6} = 2 \cdot \frac{N + 7}{2}.$$

The right side simplifies to $N + 7$. So

$$\frac{50 + N}{6} = N + 7.$$

Multiplying both sides by 6 yields $50 + N = 6N + 42$. Subtracting N from both sides yields $50 = 5N + 42$.

Subtracting 42 from both sides yields $8 = 5N$. So $N = \frac{8}{5}$.

But $\frac{8}{5} = 1.6$, which is not between 6 and 7. So there is no solution in this case.

- If $N \geq 7$, then the two middle numbers will be 7 and 7. So the median is the average which is 7. The mean must be twice the median so the mean is $2 \cdot 7 = 14$. Therefore,

$$\frac{50 + N}{6} = 14.$$

Multiplying both sides by 6 yields $50 + N = 84$. Subtracting 50 from both sides $N = 34$. This does satisfy $N \geq 7$, so the answer is 34.

※ Common Mistakes

- **Forgetting to sort before finding the median.**

The median depends on the order from smallest to largest.

- **Using the largest number as the range.**

The range is not just the biggest number. It is

$$\text{Largest} - \text{Smallest}.$$

- **Forgetting to divide when finding the mean.**

Adding the numbers only gives the sum. To find the mean, you must divide by how many numbers there are.

- **Thinking there can only be one mode.**

A list can have one mode, more than one mode, or no mode.

※ Practice Problems

1. The average of five numbers is 18. Let the first number be increased by 1, the second number by 2, the third number by 3, the fourth number by 4, and the fifth number by 5. What is the average of the set of increased numbers?
2. Two integers are inserted into the list 3, 3, 8, 11, 28 to double its range. The mode and median remain unchanged. What is the maximum possible sum of two additional numbers?
3. The numbers 3, 5, 7, a , and b have an average (arithmetic mean) of 15. What is the average of a and b ?
4. Here is a list of the numbers of fish that Tyler caught in nine outings last summer:

2, 0, 1, 3, 0, 3, 3, 1, 2.

Order the mean, median, and mode from least to greatest.

5. David writes 6 positive integers on the board (not necessarily distinct) from least to greatest. The mean of the first three numbers is 3, the median of the first four numbers is 4, the unique mode of the first five numbers is 5, and the range of all 6 numbers is 6. Find the maximum possible value of the product of David's 6 integers.
6. Maureen is keeping track of the mean of her quiz scores this semester. If Maureen scores an 11 on the next quiz, her mean will increase by 1. If she scores an 11 on each of the next three quizzes, her mean will increase by 2. What is the mean of her quiz scores currently?
7. What is the median of the following list of 4052 numbers?
$$1, 2, 3, \dots, 2026, 1^2, 2^2, 3^2, \dots, 2026^2$$
8. Camila writes down five positive integers. The unique mode of these integers is 2 greater than their median, and the median is 2 greater than their arithmetic mean. What is the least possible value for the mode?
9. Ms. Blackwell gives an exam to two classes. The mean of the scores of the students in the morning class is 84, and the afternoon class's mean score is 70. The ratio of the number of students in the morning class to the number of students in the afternoon class is $\frac{3}{4}$. What is the mean of the scores of all the students?
10. A data set consists of 6 (not distinct) positive integers: 1, 7, 5, 2, 5, and X . The average (arithmetic mean) of the 6 numbers equals a value in the data set. What is the sum of all positive values of X ?